Axiom: Do Language Models Design and Analyse Experiments Like Scientists?

A cognitive-control lab fitted to real human EEG, and a signalling lab on real single cells

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Abstract. Evaluations of AI scientists usually grade prose, often with another language model as the judge, and those judges can disagree at chance on the same outputs. We instead build a laboratory whose answers are known. Axiom-Neuro simulates a human cognitive-control study (a flanker task with working-memory load, transcranial stimulation, and EEG). Each task hides a causal mechanism among six candidate edges, and an agent must find it by designing experiments under a fixed participant-hour budget, analysing the data with standard tools, and submitting calibrated claims. Several tasks open with pilot data containing a planted confound that real human-subjects researchers fall for: a stimulation effect that is really practice, and a theta effect that is really blinks. The simulator's noise, heterogeneity and artifacts are fitted to the ERP CORE flanker dataset (40 participants) and validated on held-out participants. A second, semi-synthetic track uses real ERP CORE trials as the substrate and injects known effects on top. Scoring needs no judge: claims are compared with the hidden mechanism using a proper scoring rule. Because the same ideal analyst can be applied to any agent's data, each agent's error splits cleanly into an experiment-design part and an analysis part. A second lab, Axiom-Bio, asks the same question on 7,466 real T cells (Sachs et al., 2005): which signalling molecules directly regulate which, given real drug interventions; comparing real molecule names with abstract labels separates recalling textbook biology from inferring it from data. Hypotheses, the stopping rule and all analyses were preregistered before any language-model episode ran, and the whole study runs unattended on free infrastructure with a public results ledger. Language-model runs are in progress, and results are added to the public ledger as they complete.

Project and data: `https://github.com/sumeetmusfirah/axiom-neuro`